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RESEARCH ARTICLE

LLTQ+: A Hardware-Friendly Quantization Framework for Modern YOLO Architectures

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ABSTRACT YOLO-based object detection models are widely used in real-time applications due to their high accuracy and fast inference speed. However, their complex architectures and computational demands present challenges for deployment on low-power edge devices. To address this, we propose LLTQ+, an enhanced version of the hardware-friendly quantization method LLTQ. The proposed approach preserves batch normalization layers during Quantization-Aware Training (QAT) to maintain training stability and accuracy, and introduces a quantization strategy that preserves the representational capacity of RepConv, a key structural component of YOLO networks. Experimental results on the PASCAL VOC dataset demonstrate the effectiveness of LLTQ+. On YOLOv10-s, LLTQ+ achieved 80.6% mAP(0.5) and 61.8% mAP(0.5:0.95) under integer-only inference, surpassing LLTQ by 0.9 and 1.7 percentage points, respectively. On YOLOv9-t, LLTQ+ reached 52.9% mAP(0.5:0.95), an improvement of 0.5 points over LLTQ. Consistent performance gains were also observed across other architectures such as YOLOv7 and YOLOv7-tiny. These results confirm that LLTQ+ effectively supports integer quantization for even the latest and more complex YOLO networks, providing a practical quantization solution that balances accuracy and computational efficiency.

INDEX TERMS Computer vision, convolutional neural network, object detection, network compression, quantization.

I. INTRODUCTION

Recently, deep learning has achieved outstanding performance in a variety of domains such as computer vision, natural language processing, and speech recognition [1] and [2], driving its adoption in applications including autonomous vehicles and drones. In particular, object detection using Convolutional Neural Networks (CNNs) [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], and [13] has become a prominent research topic in computer vision, delivering high accuracy and robustness across diverse scenarios.

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Among various object detection frameworks, one of the most prominent is YOLO (You Only Look Once). Originally introduced by Redmon et al. in 2016, YOLO has undergone continuous evolution, culminating in the release of YOLOv11. Recent versions of YOLO have integrated diverse architectural modules to enhance training performance, among which RepConv (Re-parameterizable Convolution) stands out. RepConv improves the network's representational capacity during training by employing a multi-branch structure, and then re-parameterizes it into a single-path structure for inference to accelerate computation. This design has become a crucial component in boosting the efficiency and accuracy of YOLO-based object detectors.